Neural Operators for Real-Time Ultrasound Propagation

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Motivation

- **New Class (Applied DSP):** Developing an interactive, real-time simulator for students to experiment with Delay-and-Sum (DAS) beamforming and ultrasound.
- **Class Project:**
  - Students generate a TX pulse and steering vector.
  - The wave propagates through a 2D heterogeneous tissue medium.
  - The system recovers simulated RX data across a 64-element Uniform Linear Array (ULA).
  - Students recreate an Ultrasound image using DAS beamforming for real time Ultrasound video streaming.
- **Latency Issues:**
  - Target latency: ~ 100 ms for real-time interactivity.
  - Currently, PDE solvers 3–4 minutes for high accuracy simulation.

Ultrasound Propagation in a 2D Medium

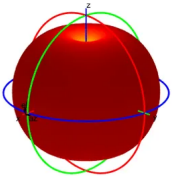
- **The imaged medium** is modeled as a static, heterogeneous 2D space $\Omega \subset \mathbb{R}^2$.
- **We can characterize the medium with 3 vector maps:**
  - $c(x, z)$: Sound speed map.
  - $\rho(x, z)$: Density map.
  - $M(x, z)$: Inclusion mask (for cysts, reflectors, and absorbing regions).
- **Acoustic Impedance** $(Z(x, z) = \rho(x, z)c(x, z))$
  - Spatial variations in impedance cause the scattering, refraction, and phase distortion that create ultrasound speckle.

Excitation Space & Transducer Geometry

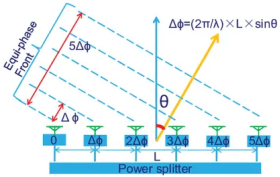
- We have a **64-element transducer** in a Uniform Linear Array (ULA) capable of both transmitting and receiving.
- **Transmit Parameters:** Defined by a transmit pulse $s(t)$ and a steering angle θ.
- **Beam Steering:** Geometric time delays applied to the i-th element define the wavefront:

$$\Delta t_i = \frac{x_i \sin\theta}{c_0}$$

Monopole antenna



Phased array built from monopoles



k-Wave Solver

- The industry gold standard PDE solver for ultrasound is **k-Wave**, a pseudo spectral PDE numerical solver
  - Evolves acoustic pressure p, particle velocity u, and density ρ through coupled nonlinear equations.
  - Must march forward sequentially in tiny time increments (Δt).
  - Time step size is strictly bounded by CFL stability conditions, making low latency execution impossible.

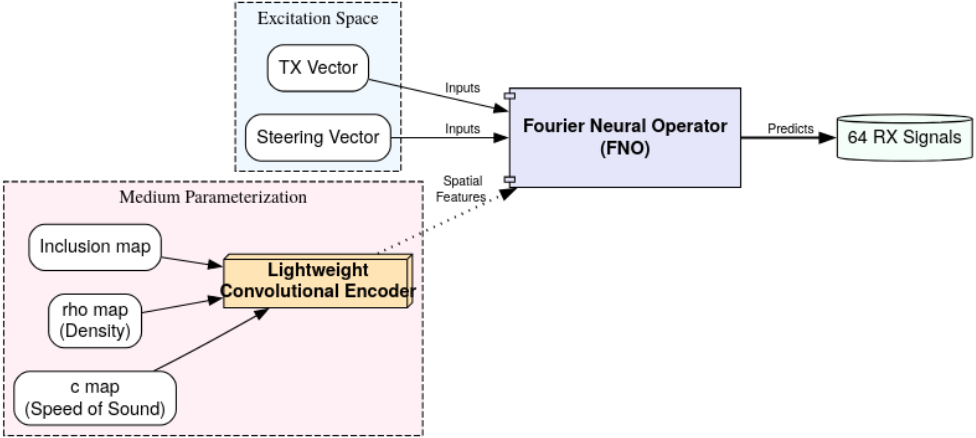
Operator Learning

- Instead of iterating through time, can we learn the forward map in a single inference pass?
- **Mapping Function Spaces:**

$$G_\theta : \mathcal{A} \times \mathcal{E} \to \mathcal{Y}$$

  - Maps the medium space $\mathcal{A}$ and excitation space $\mathcal{E}$ directly to the received RF channel signals $\mathcal{Y}$.
- We aim to **reduce latency from minutes to milliseconds** while preserving critical acoustic features.

Attempt 1: Fourier Neural Operator (FNO)



Phase Accuracy & Beamforming

- Ultrasound imaging relies on phase coherent Delay-and-Sum beamforming.
- For example, a half-wavelength error at 4 MHz is just ~ 0.2 mm in tissue (or 125 ns). However, this would give us a completely different image that a phase accurate image.
- This is because of **destructive interference.** If the phase is wrong, signals cancel each other out, and the beamformed image collapses entirely.

The k-Modes Problem

- **FNO truncation** means that the FNO nly keeps lowest k Fourier modes
- A 2MHz TX pulse carries information its higher harmonics, including 16 MHz.
- **Wave resolution constraint:**

$$\lambda = \frac{c_0}{f}, \qquad k_{\max} \sim \frac{L}{\lambda} = \frac{Lf}{c_0}$$

- Example (imaging depth $L = 4$ cm, $c_0 = 1540$ m/s):

$$k_{\max} \approx \frac{0.04 \cdot 16 \times 10^6}{1540} \approx 415$$

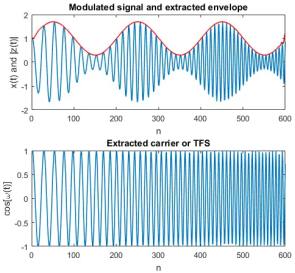
- Need $\sim 400+$ modes to resolve 16 MHz content
- Typical FNO uses ~ 32–64 modes
- $\Rightarrow$ high-frequency phase structure is discarded
- This causes **large phase errors (up to 15 degrees)** leading to beamforming collapse

IQ Modulation

- **Reducing bandwidth** by converting the raw RF signal to a complex baseband representation:

$$s_{IQ} = I + jQ$$

- Then, we **decimate**, allowing the model to train on the lower-frequency envelope, preserving the instantaneous phase while drastically reducing the required Fourier modes (16 – 32 modes).



Phase-Sensitive Loss Functions

- **MSE Loss:**

$$L_{MSE} = \frac{1}{N} \sum_{t} |\hat{y}(t) - y(t)|^2$$

- **Spectral Loss:**

$$L_{spec} = \sum_{\omega} ||\hat{Y}(\omega)| - |Y(\omega)||^2$$

- **Phase Loss:**

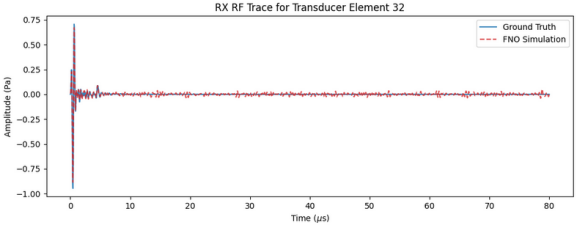
$$L_{phase} = \frac{1}{N} \sum_{\omega} |\angle \hat{Y}(\omega) - \angle Y(\omega)|$$

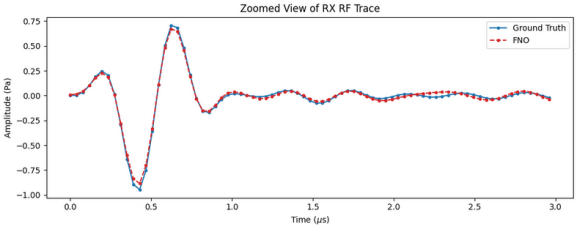
- **Amplitude-Weighted Phase Loss:**

$$L_{wphase} = \frac{\sum_{\omega} |Y(\omega)| \, |\angle \hat{Y}(\omega) - \angle Y(\omega)|}{\sum_{\omega} |Y(\omega)|}$$

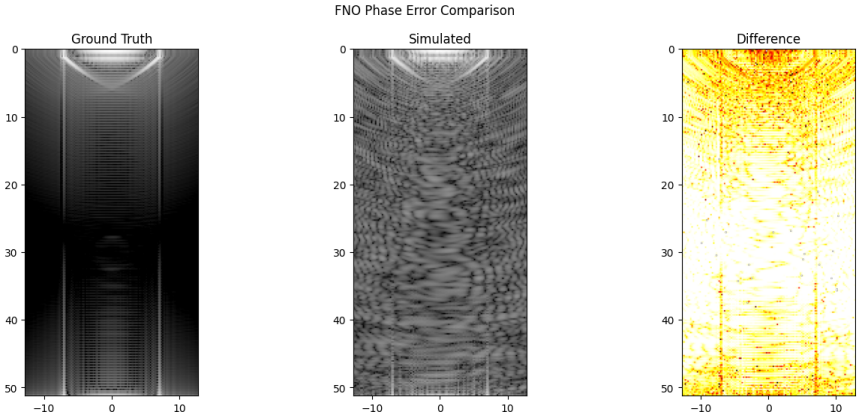
$$L = 0.8 L_{wphase} + 0.1 L_{spec} + 0.1 L_{mse}$$

FNO Results 1





FNO Results 2



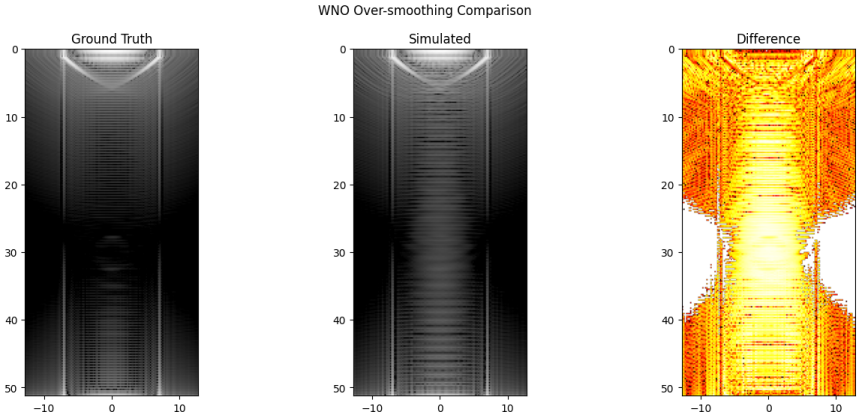
Attempt 2: Wavelet Neural Operator (WNO)

- Unlike Fourier's complex exponentials, wavelets are localized in both time and frequency:

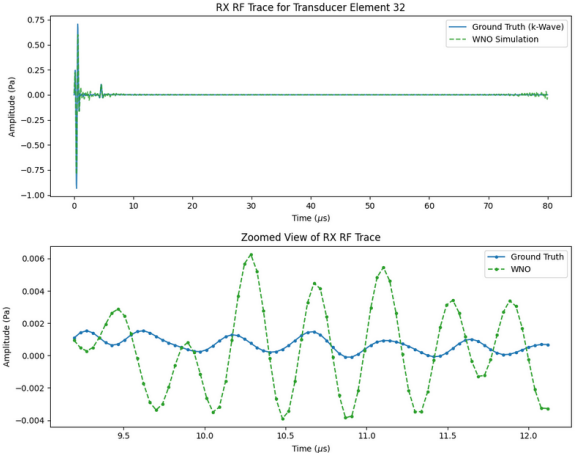
$$\mathcal{W}\{v\} = [c_A, c_D]$$

- Approximation coefficients (c_A) handle bulk wave travel and low-frequency attenuation.
- Detail coefficients (c_D) preserve sharp transient pulse edges and localized scattering events.

WNO Results 1



WNO Results 2



Conclusion

- For ultrasound propagation approximation, standard reconstruction loss (MSE) is a poor proxy for downstream imaging quality.
- The **FNO is a fantastic predictor of a time domain signal**, however, has small but noticeable phase dependencies that propitiate throughout the system and collapse the beamformed image.
- Wavelet representations align much better with the transient nature of ultrasound than Fourier representations and the **WNO results in a much higher quality beamformed image**.
- For static tissue without severe nonlinearities, the **LTI spatial interpolation lookup table approach still outperforms both neural operators**, yielding $< 0.1^\circ$ phase error.

Thank You

Questions?